Drift-free femtosecond timing synchronization of remote optical and microwave sources

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Femtosecond mode-locked lasers have revolutionized many fields of science and engineering[1–4]. Because of their ultralow noise[5], it has been anticipated that mode-locked lasers would synchronize large-scale scientific facilities[6–9] requiring extremely high timing accuracy. However, the lack of long-term stable synchronization techniques has hindered the realization of pervasive synchronization with such lasers. Here we present a comprehensive set of new techniques for long-term stable synchronization of optical and microwave sources over long distances. We use ultralow-noise optical pulse trains generated from mode-locked lasers as the timing signals, then distribute them by means of timing-stabilized fibre links and, finally, synchronize the delivered timing signals with the optical and microwave sources being targeted. Using these techniques, we demonstrate, for the first time, that remotely located lasers and microwave sources can be synchronized with less than 10-fs precision over more than 10 h. This drift-free operation is an important milestone in transitioning mode-locked laser-based synchronization from the laboratory into real-world facilities.

Advances in femtosecond mode-locked lasers have revolutionized many fields of science and engineering, and most recently by providing millions of regularly spaced optical frequency markers for time/frequency measurements[1,2] and high-precision laser spectroscopy[3,4]. One of the most fascinating aspects of mode-locked lasers is that they can simultaneously provide ultralow-noise optical and microwave signals in the form of optical pulse trains[5]. Because of this dual property of optical pulse trains, it is possible to build high-precision timing networks that tightly synchronize multiple microwave and optical sources. Modern large-scale scientific facilities such as particle accelerators, X-ray free-electron lasers (FELs)[6–8] and phased-array antennas for radio-astronomy[9] require extremely high timing accuracy between multiple remotely located microwave and optical sources, and it has been anticipated that mode-locked laser-based techniques could be used to synchronize such facilities with an unprecedented timing accuracy at the femtosecond (and even subfemtosecond) level. Although there has been remarkable progress in the development of optical atomic clocks and in transferring time/frequency standards over long distances in recent years[10–12], no previous work has demonstrated drift-free timing synchronization of optical and microwave sources over hundreds of metres with femtosecond accuracy maintained over extended periods of time (multiple hours), which is an essential prerequisite for the operation of large-scale facilities.

Here we have established a new application for mode-locked lasers by using ultralow-noise optical pulse trains for high-precision timing distribution and control of large-scale facilities. We report on the first demonstration of a set of techniques that enables large-scale synchronization with sub-10 fs timing accuracy maintained over more than 10 h. The demonstrated relative timing stabilities in timing distribution, optical–optical synchronization and optical–microwave synchronization were 2.5×10^{-20}, 9×10^{-21} and 1.9×10^{-19}, respectively, which represents many orders of magnitude improvement compared with previous work[10,13,14].

Figure 1 shows a schematic layout of the synchronization system. First, an ultralow-noise optical pulse train is generated from a mode-locked laser. By tightly locking the mode-locked laser to an optical/microwave reference, this laser serves as the optical master oscillator. The optical pulse train is then distributed to critical microwave and optical subsystems. To distribute the optical pulse train, we used dispersion-compensated and timing-stabilized fibre links. At the end-stations, tight optical–optical or optical–microwave synchronization is required. Optical–optical synchronization enables the synchronization of remotely located mode-locked lasers with the delivered pulse train. Optical–microwave synchronization synthesizes a low-noise microwave signal, the phase of which is locked to the delivered pulse train. Compared with other optical techniques relying on the optical carrier phase, such as continuous-wave transmission[15], the use of pulse trains enables simple and direct stabilization and synchronization of optical and microwave timing information through fibre links.

One of the most important requirements for high-precision synchronization is a master oscillator with low timing jitter, preferably at the subfemtosecond level in the high frequency range. This is necessary due to the limited bandwidth, at most ~100 kHz, for locking the mode-locked laser to the optical/microwave reference. The high-frequency noise beyond the locking bandwidth follows that of the free-running mode-locked laser. In addition, timing stabilization of fibre links has an intrinsic bandwidth limitation due to the link travel time of the pulse. For a 1-km-long fibre, for example, the round-trip travel time is ~10 μs. At the

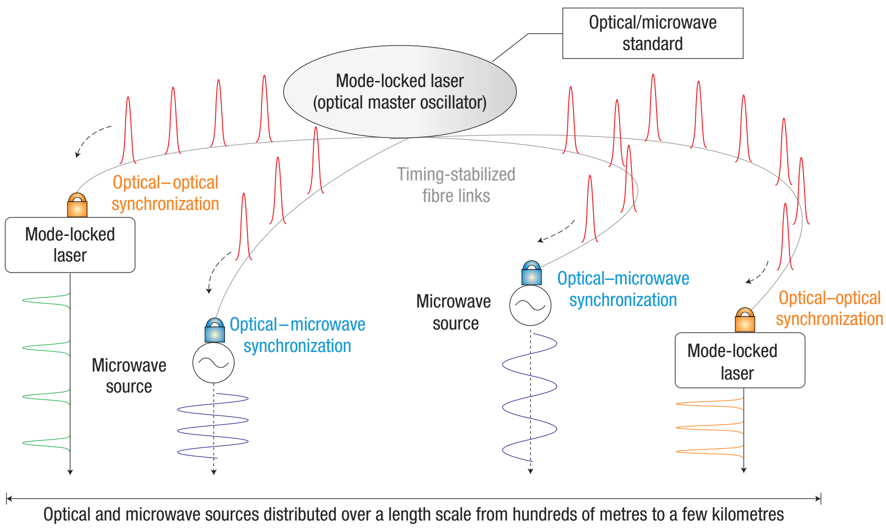

Figure 1 Large-scale synchronization of optical and microwave sources. The timing signal, a low-noise optical pulse train, is generated from the mode-locked laser (optical master oscillator). The pulse train is distributed to the optical and microwave subsystems by means of timing-stabilized fibre links. Remotely located lasers and microwave oscillators are synchronized with the delivered pulse trains, which results in pervasive synchronization of the entire system.

end-stations, timing fluctuations in the pulse train on a timescale faster than the round-trip time of the link cannot be compensated for. Recent timing jitter characterization of free-running erbium-fibre lasers confirms that the high-frequency timing jitter integrated from 100 kHz to 10 MHz is $\sim$1 fs (ref. 16), which supports the suitability of mode-locked lasers as optical master oscillators. In addition, to minimize the impact of fibre nonlinearities, a mode-locked laser with high repetition rate, for example, a few hundred MHz or higher, is desirable. Recent progress in high-repetition-rate fibre laser development has enabled mode-locked erbium-fibre lasers with a repetition rate higher than 200 MHz and less than 200 fs pulse width[17,18]. As a result, the optical master oscillator, a mode-locked laser with excellent timing jitter and high repetition rate, is readily available as a commercial product[19].

For long-distance timing distribution, the use of optical fibres is a natural choice, as it also is in optical communication networks and the dissemination of time/frequency standards[10–12]. However, acoustic noise and thermal fluctuations in the fibre cause variations in the optical path length of the fibre link. This leads to excessive noise in the arrival time of pulses transmitted over the link. For a 300-m-long fibre in a laboratory environment, this led to a timing drift of $\sim$4.5 ps in 100 s (ref. 20). Thus, the time-of-flight of each link should be stabilized with a feedback loop. The central challenge is to measure the timing drift of the fibre with high precision and thermal stability while rejecting the amplitude noise. Conventional measurement of the timing error between pulse trains is carried out by direct detection with high-speed photodiodes, filtering out a single harmonic component, and performing phase-detection with microwave mixers. However, large thermal drifts and amplitude-to-phase conversion in photodetectors and microwave mixers severely limit the performance of this method[21,22], rendering it unsuitable for long-term stable synchronization measurements.

To overcome these limitations, we developed a single-crystal balanced cross-correlator[20] to measure the timing offset between two pulse trains. This is done with an all-optical process by performing the balanced cross-correlation of a pulse back-reflected from the end of the fibre link with a new pulse entering the fibre link. Figure 2a shows the schematic of this timing stabilization. Based on the periodically poled $KTiOPO_4$ (PPKTP) single-crystal cross-correlators, we implemented two independent, timing-stabilized, $\sim$300-m-long fibre links to measure the residual synchronization error between both outputs (see Supplementary Information). Over a 72-h period of continuous, unaided operation, as shown in Fig. 2b, we observed 6.4 fs (r.m.s.) residual timing error between the link outputs, which corresponds to a timing stability of 2.5×10^{-20}. Moreover, on a short-term timescale (10 μs–1 s), we observed only 0.36 fs (r.m.s.) timing jitter.

Once the optical pulse train arrives at the end-stations, mode-locked lasers can be directly synchronized with the delivered pulse trains. Again, conventional synchronization based on photodiodes and mixers suffers from limited sensitivity and drifts. Although subfemtosecond synchronization of two lasers has been demonstrated over relatively short periods of time ($<$1 s) using photodiodes and mixers[13,23], it has not been demonstrated over extended periods of time, which is necessary for actual use in facilities.

For this optical–optical synchronization, in a manner similar to that described for the stabilization of fibre links, we can use balanced optical cross-correlation[20,24]. Figure 3a presents a generalized schematic for synchronization using the balanced optical cross-correlator. In essence, the device leverages the high timing sensitivity of nonlinear optic processes while suppressing the influence of thermal effects and amplitude noise. This is done by using temperature-insensitive time-delay elements and balanced photodetection (Fig. 3a). This scheme can be implemented for optical pulses with either the same or different centre wavelengths. In the former case, one can use the single-crystal balanced cross-correlator[20] based on birefringence as a means for introducing time delays. For the latter case, one can use a two-colour balanced cross-correlator[24] based on group-delay dispersion as a means for introducing time delays between two input pulses.

As a demonstration experiment for synchronization between two optical pulse trains from independent mode-locked lasers, we synchronized a Ti:sapphire laser and a Cr:forsterite laser using the two-colour balanced cross-correlator (see Supplementary Information). Figure 3b shows the result of the out-of-loop timing jitter measurement, performed with an out-of-loop

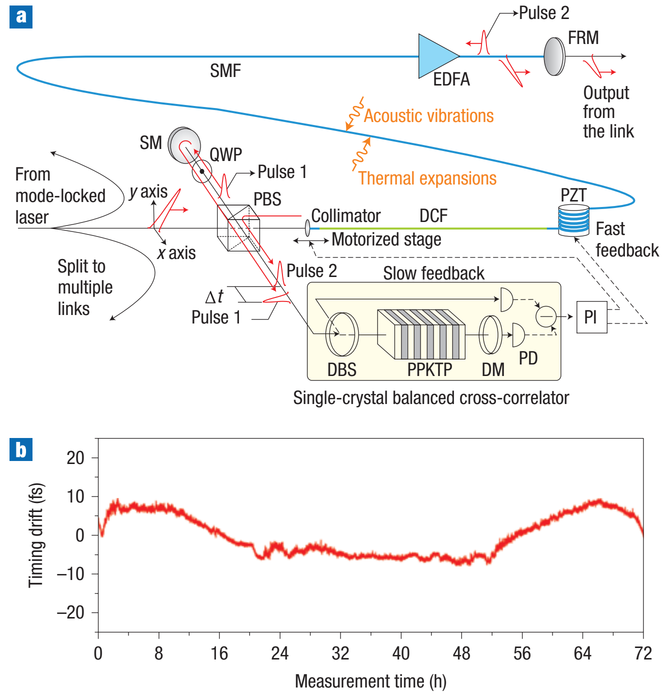

Figure 2 Timing-stabilized fibre link. a, Schematic of the timing-stabilized fibre link (SM, silver mirror; QWP, quarter-wave plate; PBS, polarizing beamsplitter; DBS, dichroic beamsplitter; DM, dichroic mirror; PPKTP, periodically poled $KTiOPO_4$; PD, photodiode; DCF, dispersion-compensating fibre; PZT, piezoelectric transducer; PI, proportional-integral controller; SMF, single-mode fibre (SMF-28); EDFA, erbium-doped fibre amplifier; FRM, Faraday rotating mirror). The input optical pulse is split into a reference pulse (pulse 1) and a pulse transmitted through the fibre link. At the fibre end, part of the fibre-transmitted pulse is coupled out as the stabilized timing signal, and the rest of the pulse is reflected back by the FRM with an orthogonal polarization state (pulse 2). Pulse 1, the reference pulse, and pulse 2, back-reflected from the link, are combined at the input of the single-crystal balanced cross-correlator. The timing offset between the two pulses, Δt, is detected by the balanced cross-correlator. The output from the cross-correlator drives the PZT and the collimator-mounted motorized translation stage to compensate the fast and slow noise introduced to the link, respectively. (See Supplementary Information for more information.) **b**, The out-of-loop balanced optical cross-correlation experimental results between the two independently stabilized 300-m-long fibre links. The data are sampled every second with a bandwidth of 100 Hz. The total timing drift integrated over 72 h is 6.4 fs (r.m.s.).

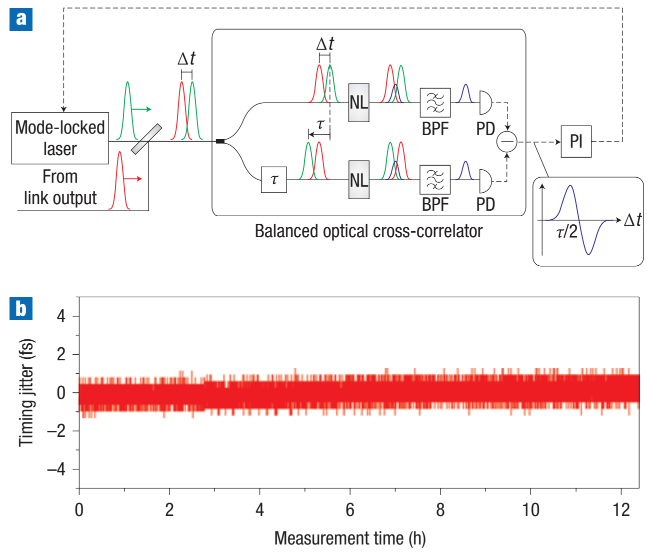

Figure 3 Optical–optical synchronization. a, Schematic of the optical–optical synchronization using a balanced optical cross-correlator (NL, nonlinear crystal; BPF, bandpass optical filter; PD, photodiode; PI, proportional-integral controller; τ, time delay element for shifting the timing offset between the two pulses). Two pulses from the fibre-link output and the mode-locked laser (which we aim to synchronize) are spatially and temporally combined at the beamsplitter, and then applied to the balanced cross-correlator. In the cross-correlator, the combined input pulses (with a timing offset of Δt) are split into two almost identical beam paths, differing only by a fixed amount of time delay (τ) between the two pulses. The nonlinear (for example, sum-frequency) components are generated at the NLs. The resulting output from the cross-correlator is shown in the inset, as a function of Δt. When the feedback loop is closed by applying this error signal to the PZTs in the mode-locked laser, the cross-correlator output is locked to the zero-crossing, where the amplitude noise is cancelled. **b**, The result for synchronization of a Ti:sapphire laser and a Cr:forsterite laser with a two-colour balanced cross-correlator. The out-of-loop optical cross-correlation is measured with a 2.3 MHz bandwidth over more than 12 h. The integrated timing jitter is 0.4 fs (r.m.s.) in the 22 µHz–2.3 MHz range.

optical cross-correlator. The integrated timing jitter over 12 h is <0.4 fs (r.m.s.) in 2.3 MHz bandwidth, which corresponds to synchronization with a relative timing stability of 9×10^{-21}.

Femtosecond synchronization is necessary not only for ultrafast lasers but also for microwave sources. For example, multiple microwave sources should be tightly synchronized with each other to generate electron beams with higher timing accuracy in accelerators and FELs.

It has been shown that the timing stability of optical pulse trains is extremely high, and that microwave signals with $\sim 1 \times 10^{-18}$ stability can be extracted from the pulse train[14]. Thus, the extraction of microwave signals from the delivered pulse trains is a promising way to generate high-quality microwave signals. However, the transfer of timing stability from the optical domain to the electronic domain has turned out to be problematic. Again, the conventional way of extracting microwave signals by direct photodetection cannot deliver the needed stability. Although subfemtosecond short-term (<1 s) jitter has been demonstrated[25], it is very difficult to keep this level of precision over extended periods of time, due to timing drift in the photodetection[14,21,26]. So far, the best frequency stability obtained with conventional photodetection is 2.6×10^{-17}, which is one order of magnitude worse than the performance projected from the optical stability of the pulse train[14].

To achieve full performance of timing extraction, we have implemented an optoelectronic phase-locked loop (Fig. 4a) that locks the zero-crossings of a microwave signal to the optical pulse train. The key issue in implementing the phase-locked loop is detecting the phase error between the optical pulse train and the microwave signal in the optical domain before the photodetection is involved. For this, we convert the phase error between the pulse train and the microwave signal into an optical quantity, such as intensity imbalance[27] or intensity modulation depth[28], which is robust against photodetection excess noise. In this way, we can extract an error signal in the electronic domain that is not corrupted by the excess noise. This clean error signal is then used to drive a high-quality voltage-controlled oscillator. Here we implemented this technique by using a recently demonstrated optoelectronic device, a balanced optical–microwave phase detector (BOM-PD), based on electro-optic sampling with a

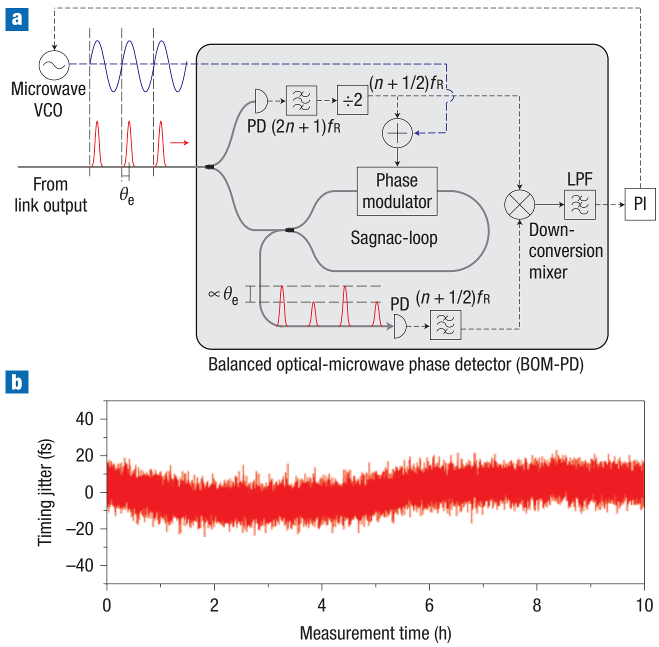

Figure 4 Optical–microwave synchronization. a, Schematic of the optical–microwave synchronization using a balanced optical–microwave phase detector (BOM-PD; VCO, voltage-controlled oscillator; PD, photodiode; LPF, low-pass filter; PI, proportional-integral controller; f_R, pulse repetition rate). The optical pulse train from the fibre-link output and the microwave signal from the VCO (which we aim to synchronize) are applied to the BOM-PD. Part of the pulse train is tapped off to generate a reference signal ($(n+1/2)f_R$), which is used both for differential biasing of the Sagnac loop and for synchronous detection. The rest of the pulse train is applied to the Sagnac loop with a phase modulator driven by the sum of the microwave signal (to be locked) and the reference signal. At the output of the Sagnac loop, the output pulse train is modulated with an amplitude proportional to the phase error (θ_e) between the pulse train and the microwave signal. The amplitude modulation depth is then converted to the baseband error signal by a synchronous detection using a down-conversion mixer. When the feedback loop is closed by applying this error signal to the VCO, the zero-crossings of the microwave signal are locked to the pulse train. **b**, The measurement result of the out-of-loop relative timing jitter between the 200.5 MHz optical pulse trains and the regenerated 10.225 GHz (the 51st harmonic) microwave signal over 10 h. The integrated timing jitter is 6.8 fs (r.m.s.) in the 28 μHz–1 MHz range.

differentially-biased Sagnac-loop interferometer[28,29]. Figure 4a shows the schematic of optical–microwave synchronization based on the BOM-PD.

Figure 4b shows the relative timing jitter measured over 10 h. One BOM-PD is used to regenerate a 10.225 GHz microwave signal from a 200.5 MHz optical pulse train. The other BOM-PD is used for monitoring the out-of-loop timing jitter between the locked microwave signal and the optical pulse train (see Supplementary Information). The measurement shows that the timing jitter in a 1 MHz bandwidth integrated over 10 h is 6.8 fs (r.m.s.). This corresponds to 1.9×10^{-19} relative timing stability of the microwave signal, which is, to our knowledge, the highest stability achieved for a microwave signal readout from mode-locked lasers.

In summary, a set of techniques for achieving drift-free sub-10 fs timing accuracy between remotely located optical and microwave sources over multi-hour continuous operating periods has been demonstrated. The demonstrated capability for high-precision, drift-free synchronization is critical to moving forward with the design and operation of high-precision facilities and networks such as EUV/X-ray FELs, large-scale accelerators, radio-astronomy systems and future communication networks.

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